India’s Strategy for AI Leadership

A Ground-Up Five-Layered Approach

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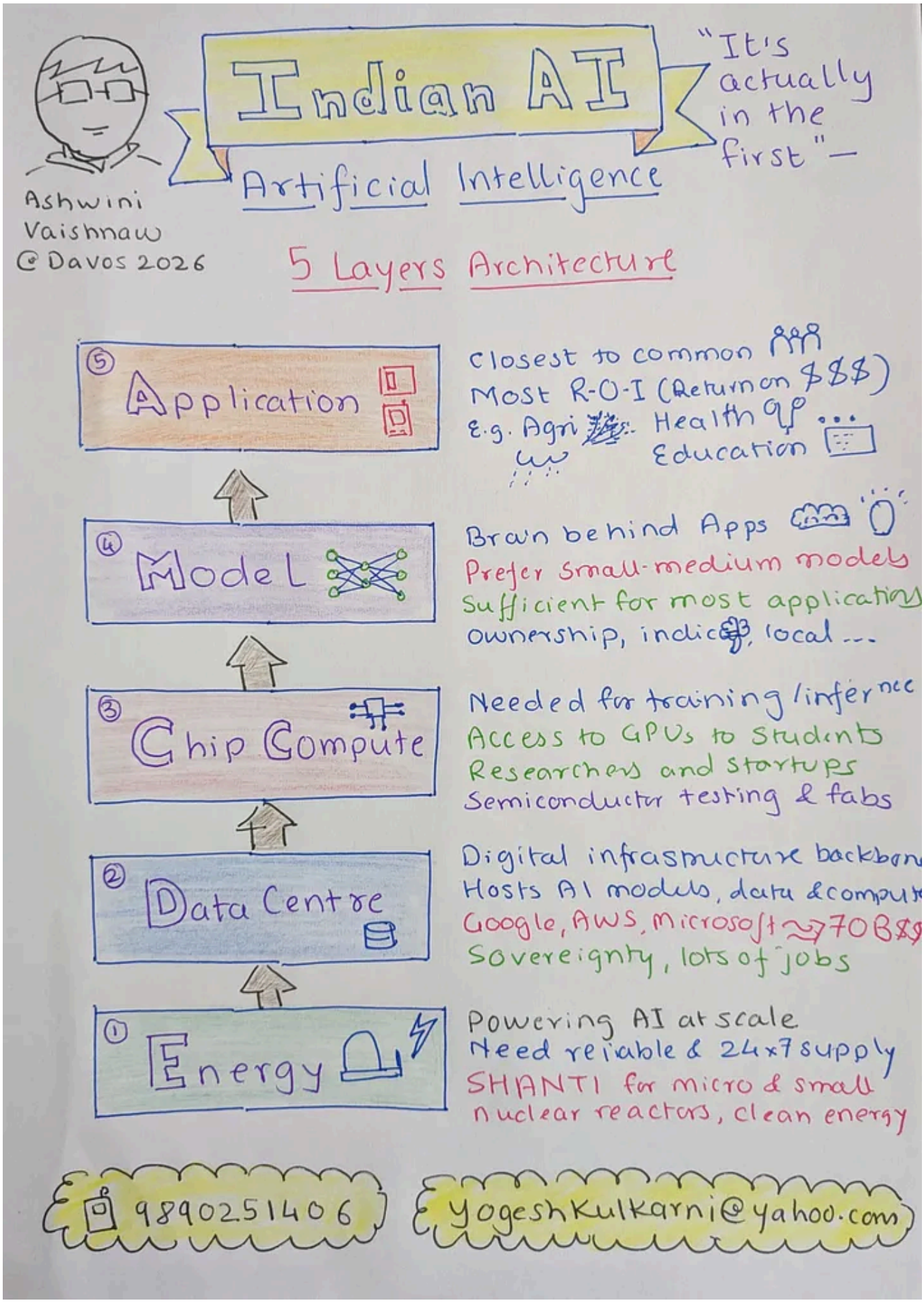

(Sketchnote by the author)

When India's Minister for Electronics and Information Technology Ashwini Vaishnaw took the stage at the World Economic Forum in Davos, he made a bold claim that turned heads. While some observers had placed India in the second tier of AI nations, Vaishnaw insisted his country belongs firmly in the first group. His reasoning reveals a sophisticated understanding of how artificial intelligence will actually reshape the global economy.

At the heart of India's approach is a five-layer framework that Vaishnaw calls the AI architecture. Think of it like building a house. You need a solid foundation, strong walls, reliable utilities, and functional rooms. Similarly, AI needs five essential layers working together, and India is investing in all of them simultaneously.

The first layer is applications, the part of AI that everyday people actually see and use. This is where farmers get crop advice, students access personalized learning, or hospitals diagnose diseases faster. India is betting big here. The country aims to become the world's largest supplier of AI services. The strategy is straightforward: Indian companies will go into businesses around the world, understand how they work, and provide custom AI solutions that deliver real returns on investment. This matters because, as Vaishnaw points out, actual business value doesn't come from building the biggest AI system. It comes from solving real problems efficiently.

The second layer is models, the brains behind AI applications. While companies like OpenAI and Google have grabbed headlines with massive AI models containing hundreds of billions of parameters, India is taking a different path. The country is building what Vaishnaw calls a bouquet of smaller models, typically with twenty to fifty billion parameters. These models can handle ninety-five percent of the work most organizations need, and they're already being deployed across multiple sectors to boost productivity. India has made these models freely available to its population, ensuring that startups, researchers, and students all have access to the tools they need.

The third layer is chips and computing power. Here's where things get expensive. The powerful graphics processing units needed to run AI cost a fortune, and in wealthy nations, big technology companies often control access to them. India chose a different approach through public-private partnerships. The government has enabled access to thirty-eight thousand GPUs at approximately

one-third the cost found in most countries. This democratizes AI development, ensuring that a bright student in Bangalore has the same access to computing power as researchers in Silicon Valley.

The fourth layer is data centers, the physical infrastructure that houses all this technology. These facilities store AI models, data, and computing resources. India has attracted massive investments here, with technology giants like Google, Microsoft, and Amazon committing seventy billion dollars to expand data center capacity in the country. These facilities are becoming more efficient too, with innovations in cooling systems and energy use. Beyond the technology, data centers strengthen what Vaishnaw calls digital sovereignty, keeping Indian data under Indian control while creating high-value jobs.

The fifth and final layer is energy. This is the foundation that makes everything else possible. AI systems are power-hungry, requiring electricity around the clock. As India's AI infrastructure grows, so do its energy needs. Solar and wind power help, but they're intermittent. The sun doesn't always shine, and the wind doesn't always blow. For AI systems that can't afford downtime, India is turning to nuclear energy through the recently proposed SHANTI Act. Small modular and micro reactors would provide clean, stable power specifically for AI infrastructure through public-private partnerships and foreign investment.

What makes India's strategy particularly interesting is its focus on economics over prestige. While other nations race to build the largest AI models, Vaishnaw questions whether that even creates geopolitical power. He suggests that companies spending billions on massive models might actually go bankrupt in coming years because the economics don't work. A fifty-billion parameter model running on one GPU, or even a thirty-billion parameter model that doesn't need a GPU at all, can deliver better returns for most real-world applications.

This practical approach extends to how India is preparing its workforce. The country is training ten million people in AI skills, ensuring that its large information technology industry can pivot toward providing AI-enabled services both domestically and globally. The goal is AI diffusion at population scale, spreading the technology's benefits widely rather than concentrating them in a few elite institutions or companies.

India is also taking what Vaishnaw calls a techno-legal approach to AI governance. Instead of just passing laws and hoping companies comply, the government is developing technical tools to address AI’s challenges. These include technologies to detect and counter bias in AI systems, identify deepfakes with court-ready accuracy, and ensure proper data handling before models are deployed in businesses.

According to Stanford’s rankings, India ranks third globally in AI penetration and second in AI preparedness and talent. For a country that only recently became an active geopolitical player, this represents remarkable progress. The five-layer strategy shows that success in AI isn’t just about having the biggest models or the most powerful chips. It’s about building a complete ecosystem where applications deliver value, infrastructure is accessible, and the whole population can participate in what Vaishnaw calls the fifth industrial revolution.

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